Poker Assignment Breakdown

Through the rest of this course, you will be doing the Poker Monte Carlo Simulation Project (or "Poker Project"). This project is going to be a lot bigger than what you have written so far, so we will break it down into 9 different assignments. These 9 assignments will bring together many of the programming skills you have learned with us so far.

The first assignment will be to make a Card class, so that we can represent one card. The card will have a suit and a value, and you will write a constructor and some methods for it.

In the second assignment, you will make a Deck class to represent a deck (or also a hand) of cards. In the third, you will read input from a file to create Decks. These three assignments will use a lot of Python skills from the first three courses, such as lists, classes, and file input.

After that, you will write some test cases for poker hand evaluation—figuring out what poker hand a list of cards represents. If you aren't familiar with the rules of poker, that is ok: we'll give you a reading that will describe what you need (and we aren't doing any betting, so we don't need to worry about those rules). You'll put your test case writing skills to use to come up with 11 test cases relevant to poker hand evaluation. Writing these test cases before you write hand evaluations will help you think about some tricky cases in advance, and hopefully avoid making those mistakes. Once you have written these test cases, you will write the logic to do poker hand evaluation (don't worry—we'll break that down for you and guide you through it).

The next two assignments will involve computing the possible outcomes for an individual hand in isolation. One of these will be a Monte Carlo simulation where you know N cards, and need to draw M more cards, then evaluate the (N+M) card hand. You'll count how many times you get each outcome, and produce a table of the results. After that, you will put your Pandas knowledge to use to read in the output of the Monte Carlo simulation, and estimate probabilities if you have a subset of the cards in the tabular data. For example, if the tabular data from the Monte Carlo simulation was done on 4 card hands, but you only know three cards, you can find the rows in the table with the cards you do have, and use those to estimate your probabilities. This will be a great reminder of the Pandas skills you learned before!

Finally, we will finish our Poker Monte Carlo simulation in two more steps. One of these will be to make changes to some of your previous code to support unknown cards, which will be drawn in the future, and may be shared between multiple hands. Finally, you'll put it all together in a Monte Carlo simulation that reads a description of poker hands which may share unknown cards, draws the unknown cards from the remaining deck, evaluates the hands, and determines who wins. Repeating this process many times gives you a great estimate of the probability of each hand winning.

This project will not only let you practice a wide variety of Python and programming skills you have learned, but also let you see how breaking down a very big task into small pieces makes the task manageable.

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